NIMBioS
National Institute for Mathematical and Biological Synthesis

Application of a Sigmoid Model Fitting Algorithm to Time Course RNA-Seq Data in *Escherichia coli* (*E. coli*)

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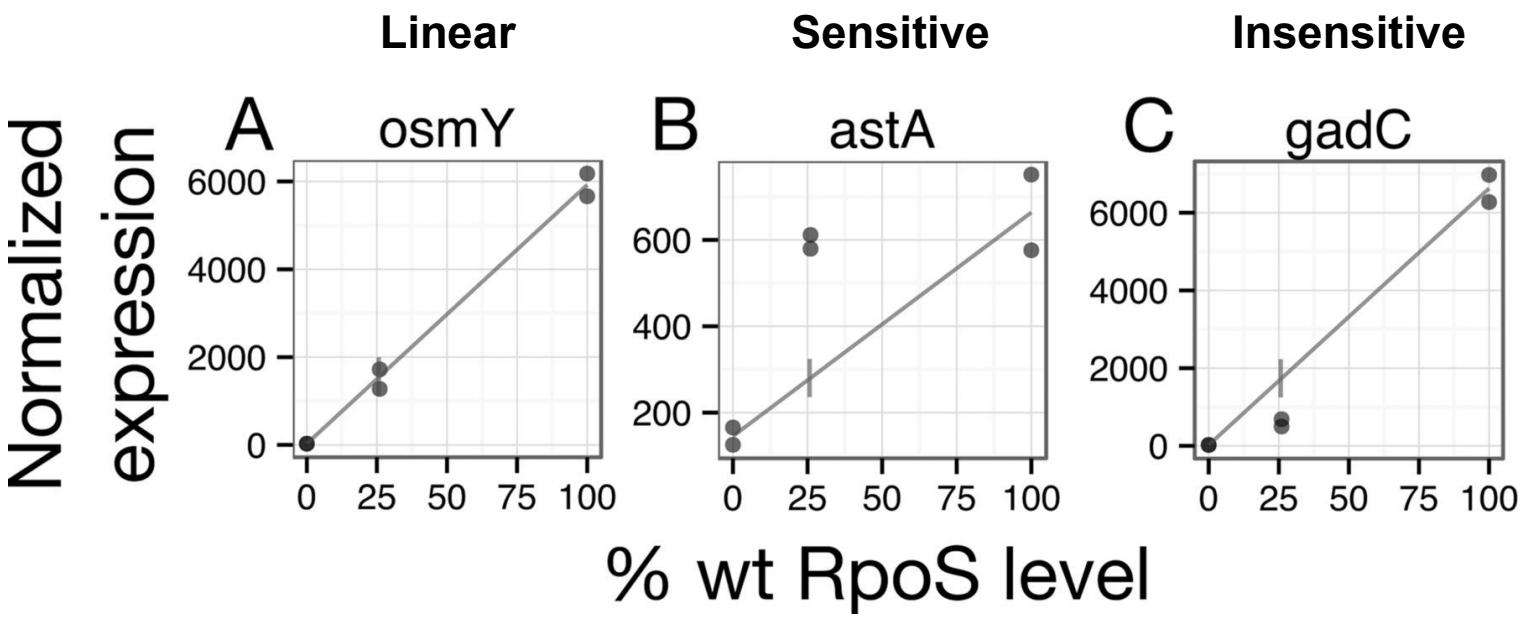
Genes in the RpoS regulon display three classes of responses



Analysis of RNA-Seq data enables inference on expression timing

The Hypothesis

Sensitive genes are induced sooner than insensitive genes.

The Data

RNA-seq data collected over 150 minutes from *E. coli* in stationary phase and osmotic shock.

Our Approach

1. Identify differentially expressed genes
2. Fit a model to expression profiles
3. Extract onset time parameter & test hypotheses

DESeq2 addresses the theoretical and practical challenges of modeling RNA-Seq data

Non-normality & dependence of variance on mean

Technical artifacts

Small # of biological replicates inhibit inference & data summarization

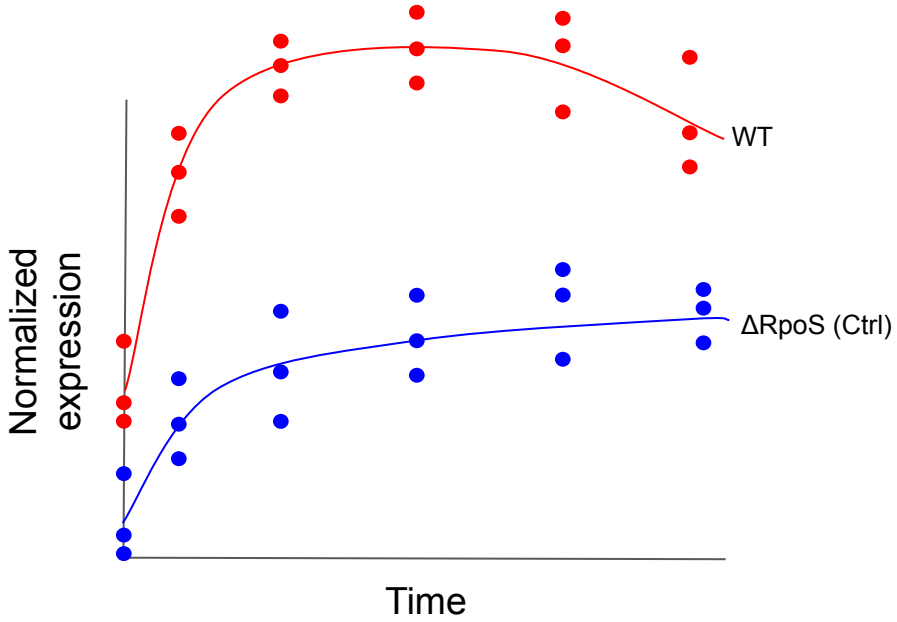
Models read counts of gene *i* in sample *j* as $K_{ij} \sim NB(\mu_{ij}, \alpha_i)$, where $Var(K_{ij}) = \mu_{ij} + \alpha_i(\mu_{ij})^2$

Scales mean reads by sample-specific normalization factors to correct for technical/sequencing artifacts.

Shrinks dispersion outliers to average dispersions of genes with similar expression. Shrinks LFC to 0 when counts are low.

DESeq2 GLM framework models experimental design

Generalized linear models (GLMs) accommodate non-normal error structure, dependence of variance on mean, and allow flexibility to analyze complex designs



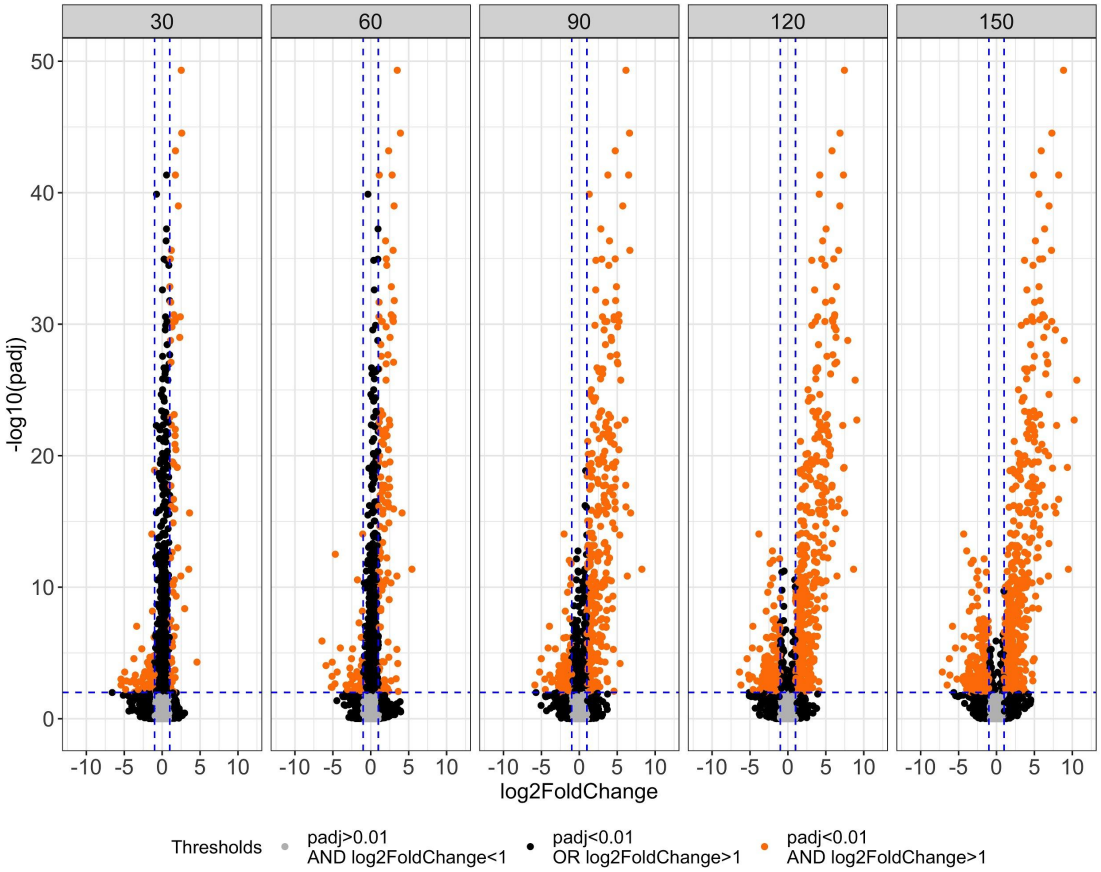
Reduced Model: Condition+Time

Full Model: Condition+Time+**Condition:Time**

LRT

- tests for differential expression kinetics by comparing full and reduced models.
- Mitigates multiple testing

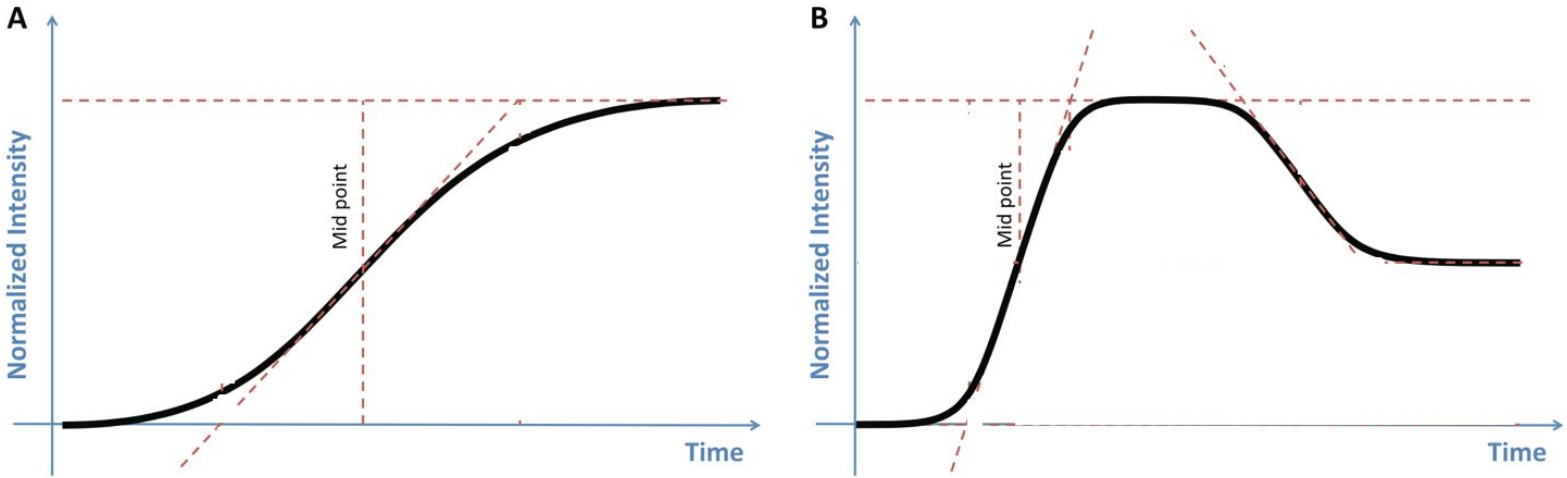
DESeq2 identifies DEGs in *E. coli* RNA-Seq data



As *E. coli* enters stationary phase, the bacterium undergoes an *asymmetric upregulatory differential expression response.*

Applying DESeq2 to stationary phase time course yields a list of **745 DEGs** (padj<0.01).

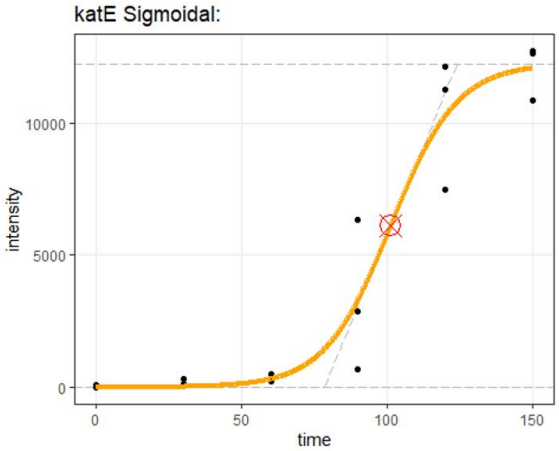
Sicegar enables sigmoid modeling of intensity data

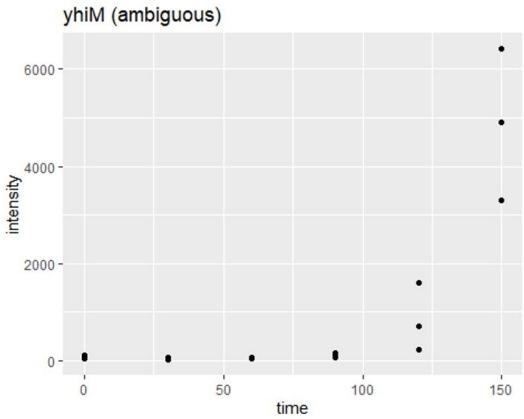


- **normalizes** the data to a [0,1] scale on x (time) and y (expression) axes.
- **minimizes** the sum of squared residuals to fit parameters
- **fits robustly** by running the fitting procedure multiple times with different parameter initializations

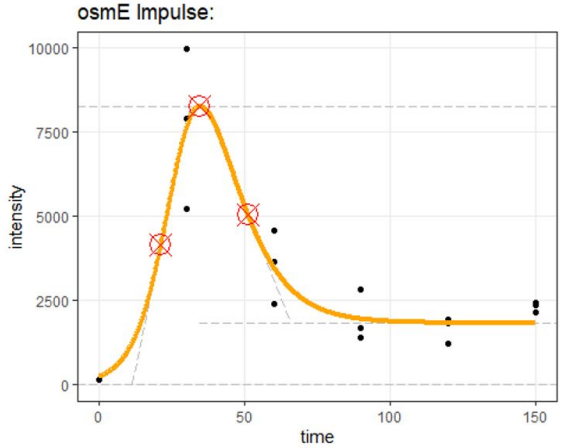
Sicegar reliably fits and categorizes RNA-Seq Data

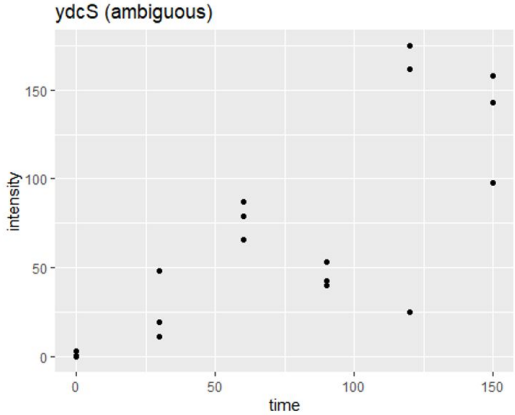
Stationary Phase



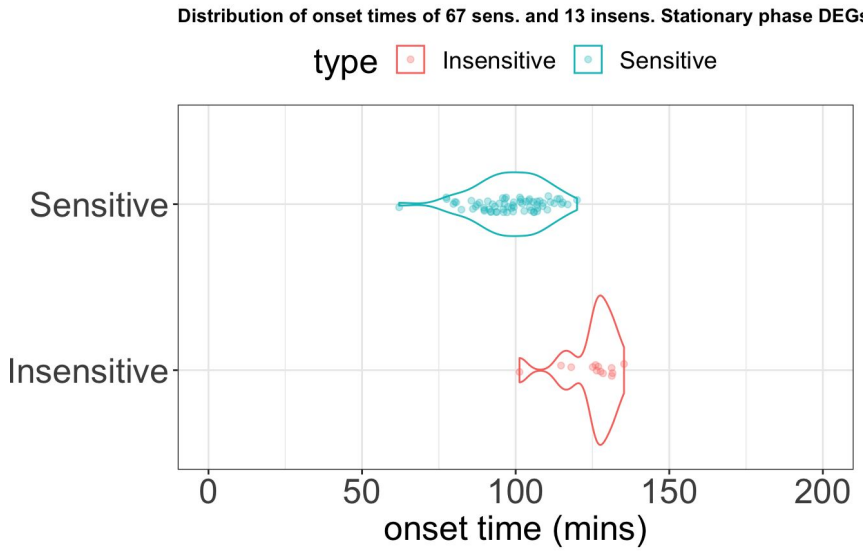


Osmotic Shock

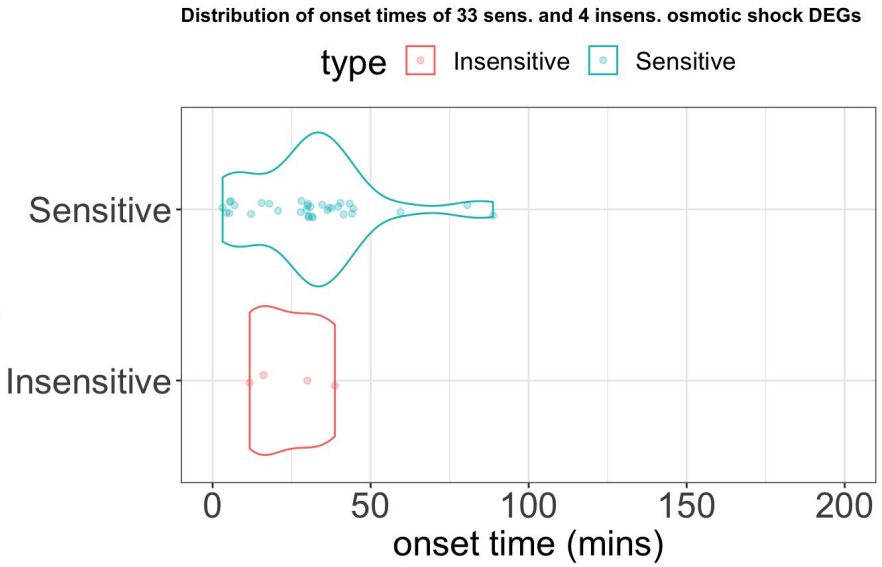




There is a significant difference in the onset times of sensitive and insensitive genes in stationary phase

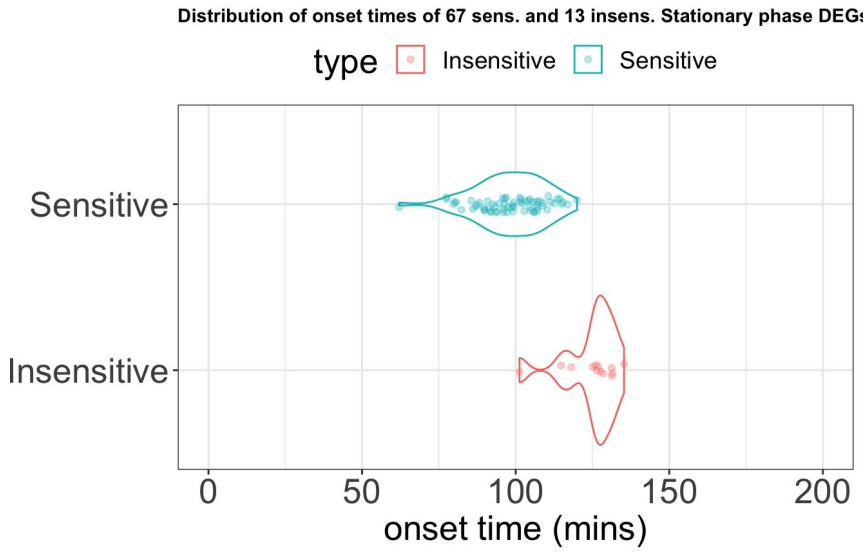


two -sample t-test: $p = 6.41e^{-12}$
95% confidence interval for difference in means: (20.04, 33.13)



two -sample t-test: $p = 0.49$
95% confidence interval for difference in means: (-13.65, 27.58)

There is a significant difference in the onset times of sensitive and insensitive genes in stationary phase



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Acknowledgements:

- Professor Daniel Stoebel
- Professor Johanna Hardin

Questions?

References

1. Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **15**, 1–21 (2014).

2. Caglar, M. U., Teufel, A. I. & Wilke, C. O. Sicegar : R package for sigmoidal and double-sigmoidal curve fitting. *PeerJ* **6**, (2018).

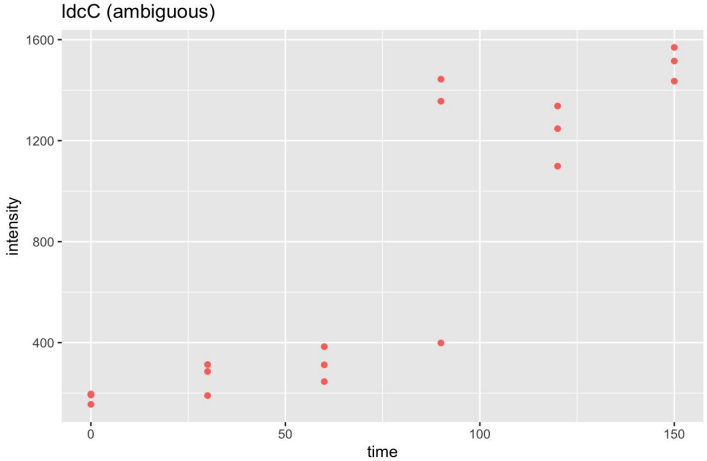
3. Wong, G. T. *et al.* Genome-Wide Transcriptional Response to Varying RpoS Levels in Escherichia coli K-12. *Journal of Bacteriology* **199**, (2017).

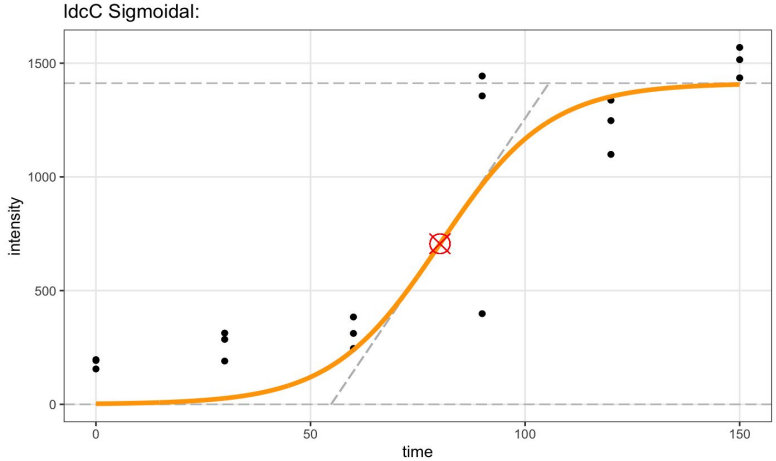
4. Beeler, S. M., Doering, C. R., Tran, S. & Stoebel, D. M. *The transcriptional response of genes to RpoS concentration in* Escherichia coli *is not determined by core promoter sequences*. http://biorxiv.org/lookup/doi/10.1101/796656 (2019) doi:10.1101/796656.

Sicegar offers customizable categorization thresholds

AIC score
- Akaike Information Criterion
- Indicates quality of fit

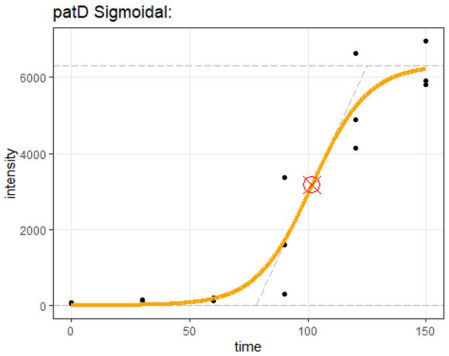
Increasing the AIC score threshold allows for the fitting of a sigmoidal model to noisier expression profiles





Threshold AIC Score: -10 0

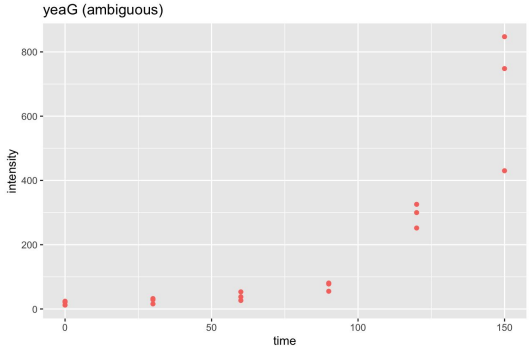
Sicegar offers customizable categorization thresholds

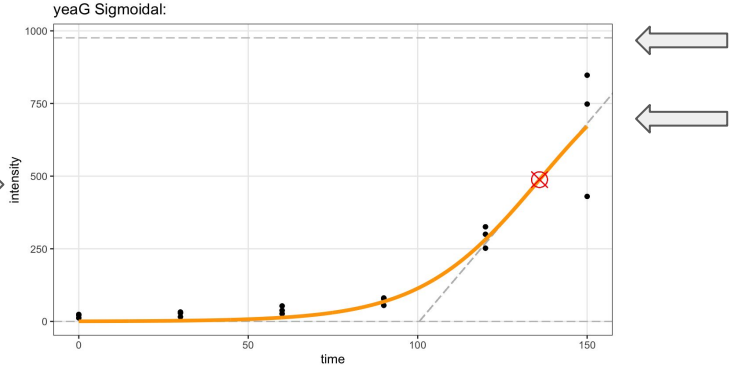


Intensity ratio

- Ratio of model at last time point to predicted max value
- *patD* intensity ratio=0.98

Decreasing the threshold intensity ratio allows fitting to profiles that have not reached a steady-state level





Threshold intensity ratio: 0.75 0.65